# THE ECONOMIC IMPACT OF COVID-19 NON-PHARMACEUTICAL INTERVENTIONS AND THE INFLUENCE OF POVERTY AND INEQUALITY

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# INTRODUCTION

- COVID-19 has spread rapidly across the world, resulting in tremendous economic damage.
- Non-Pharmaceutical Interventions (NPI) aimed to mitigate the spread of SARS-CoV-2. Impacts in both directions
- The adverse economic effects of COVID-19 may vary by the stringency of the NPIs, their length of implementation, and the degree of compliance.
- Robust evidence on how COVID-19 impacted the global economy is needed to inform decision making.

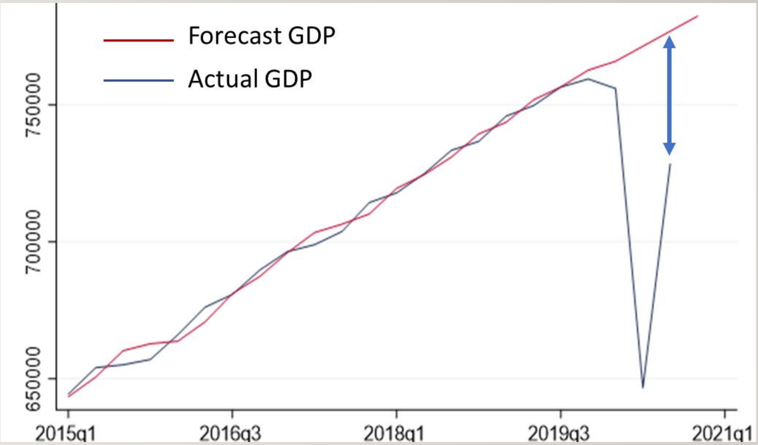
# OBJECTIVE

This work aims to estimate the variation on quarter Gross Domestic Product (GDP) and its association with the rigor of Non-Pharmaceutical Interventions (NPI´s).

# METHODS

**For 34 countries members and 8 partners countries of the Organization for Economic Co-operation and Development (OECD)**

- We gathered information on quarterly growth rates of real GDP from the OECD databases. (1)
- Additionally, we estimate projections of GDP growth using data from 2015 to 2019 using Holt Winters forecast model. (1)
- We estimated the GDP differences between forecasted and actual GDP




1. Quarterly National Accounts [Internet]. [cited 2021 Jan 9]. Available from: https://stats.oecd.org/

# METHODS

- We also estimated averages of NPI stringency and Mobility reduction observed in the 2020 quarters, using data published from the Oxford Covid-19 Government Response Tracker (2) and Google's Community Mobility Reports, respectively (3).
- Demographic and economic indicators used for controlling the econometric models were obtained from the World Bank (4).

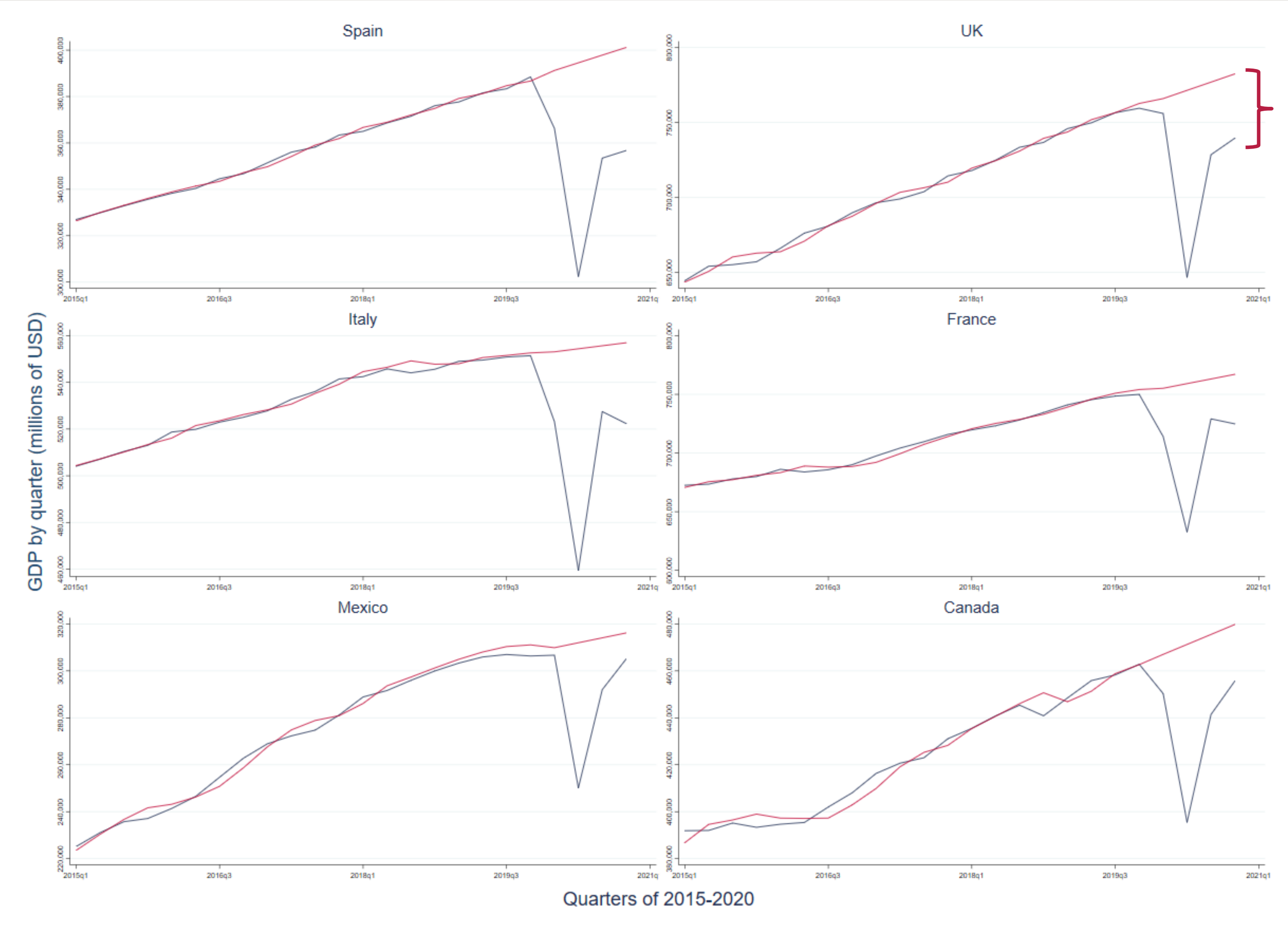
**We used a panel data model with random effects to identify the effect of the NPIs stringency attained over the GDP.**

$$GDP\ growth_{\ p,t} = \beta_0 + \beta_1 \left( NPIstringency_{p,t} * Mobility_{p,t} \right) + X\beta + \varepsilon_{p,t}$$

2. Coronavirus Government Response Tracker | Blavatnik School of Government [Internet]. [cited 2020 Sep 4]. Available from: https://www.bsg.ox.ac.uk/research/research-projects/coronavirus-government-response-tracker
3. Google. Informes de Movilidad Local sobre el COVID-19 [Internet]. [cited 2020 Sep 4]. Available from: https://www.google.com/covid19/mobility/
4. The World Bank. World Development Indicators | DataBank [Internet]. [cited 2020 Sep 4]. Available from: https://databank.worldbank.org/source/world-development-indicators

# RESULTS

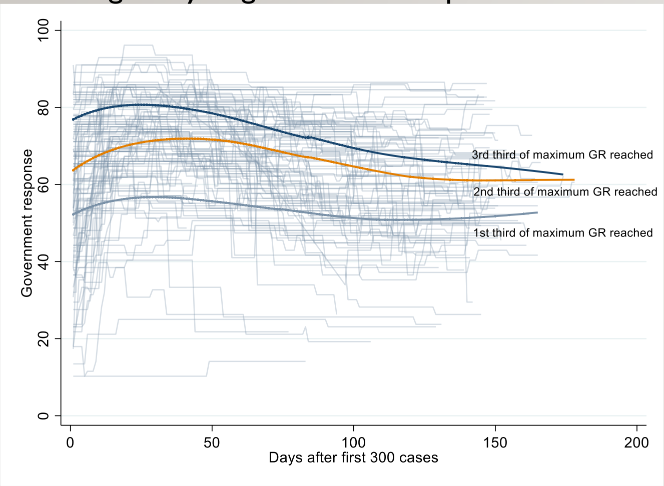
# FORECASTED VS. ACTUAL GDP



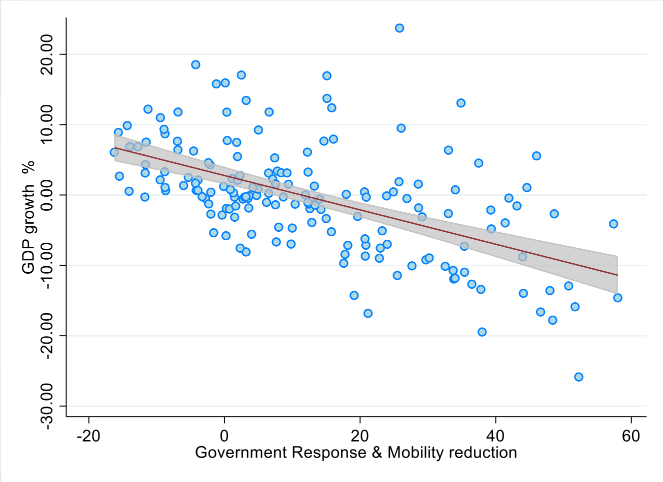

GDP difference between
Forecast and actual GDP

# RESULTS: VARIABLILITY IN GOVERNMENT RESPONSE

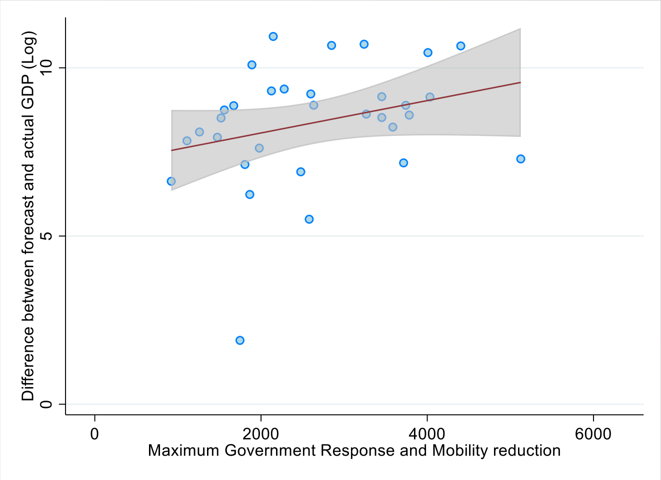
## Heterogeneity in government response



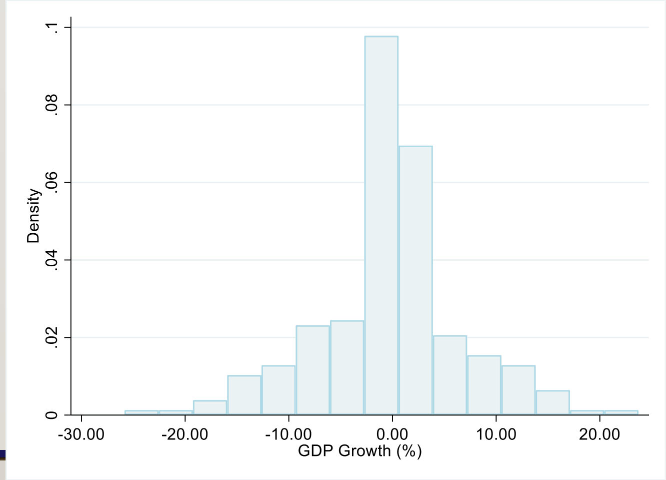

## GDP Growth vs Government response & Mobility reduction



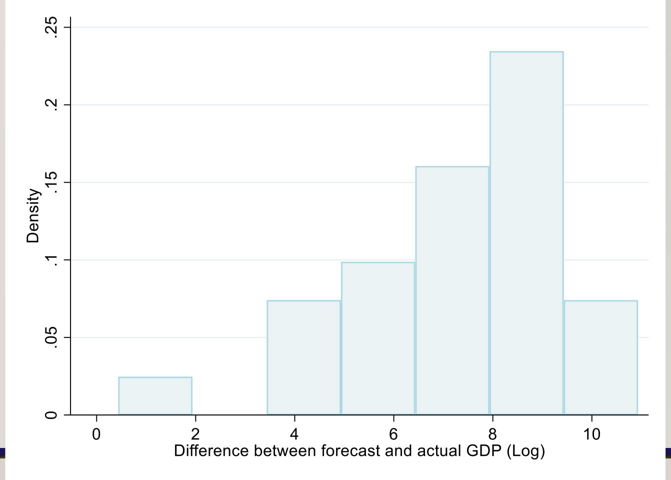

## Differences between forecasted and actual GDP vs. Maximum Government response & Mobility reduction reached



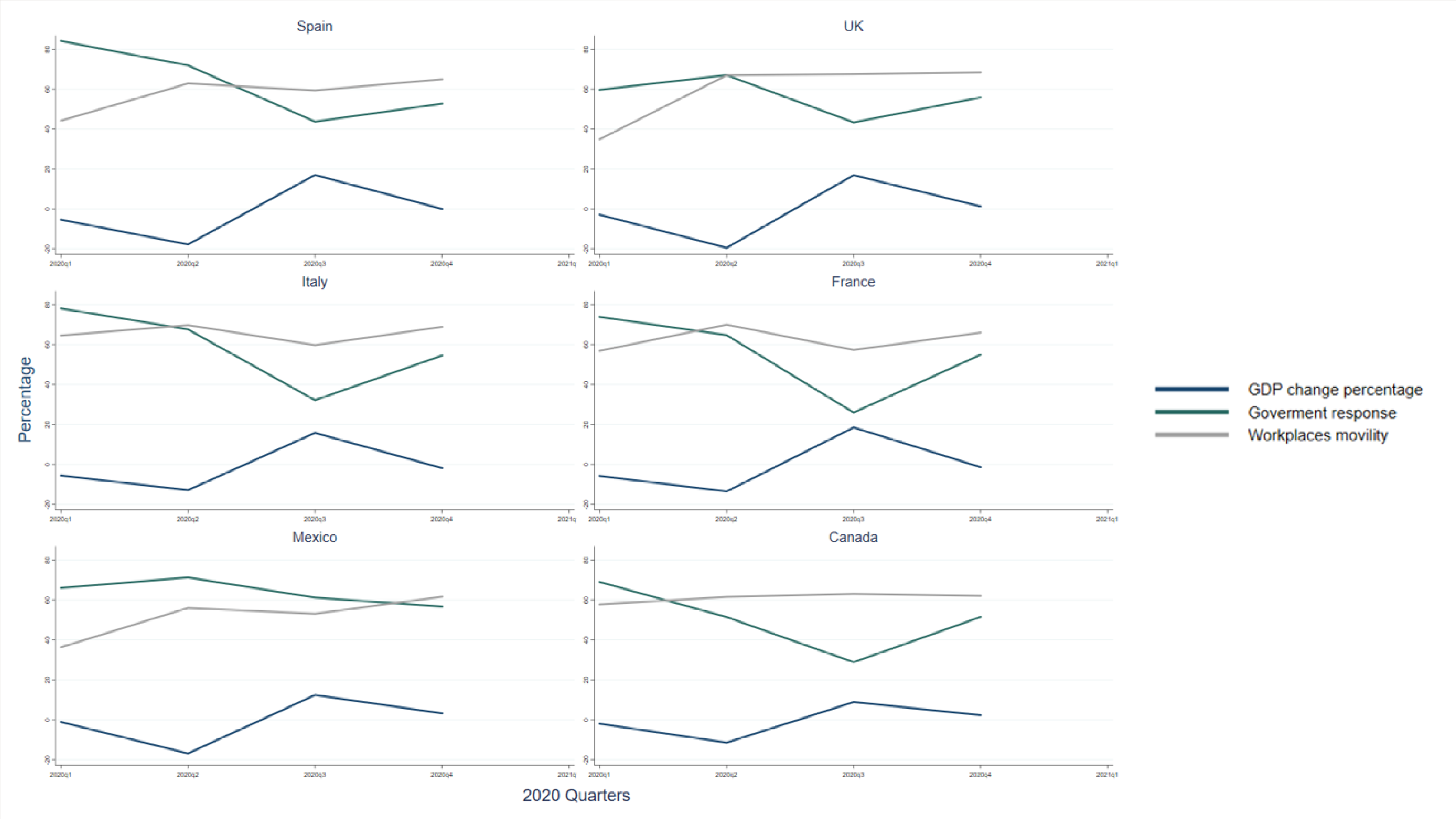

## GDP Growth in 2020 distribution




## Distribution of differences between forecasted and actual GDP



# GDP GROWTH, GOVERNMENT RESPONSE AND MOBILITY REDUCTION



| VARIABLES | GDP growth | GDP growth | GDP growth | GDP Difference |
| --- | --- | --- | --- | --- |
| Mobility reduction | -0.179*** | | | |
| | (0.0191) | | | |
| Governemnt response index | | -0.162*** | | |
| | | (0.0592) | | |
| Government response * Mobility reduction | | | -0.274*** | 283.7*** |
| | | | (0.0298) | (90.85) |
| Observations | 166 | 166 | 166 | 132 |
| Number of country_code | 42 | 42 | 42 | 34 |

Standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1
Notes: All model controlled by Population, time to preparedness since the first case, Gini index, and average of years of education

# DISCUSSION AND CONCLUSIONS

- Our estimations suggest that containment measures have had, on average, a negative impact on the economic activity
- Reduction on average of about 0.27% percent in the GDP for each additional percentage of rigor in the government response
- We estimate an association between the maximum rigor in the government response reached and the differences observed on predicted GDP at the end of 2020 vs. the observed GDP on average of 283 million dollars for each additional percentage of rigor reached.
- The estimations were consistent after controlling for other factors such as inequality -GINI index, extreme poverty, time to preparedness, and education years in the population.
- Mobility reduction showed to be an adequate modulator of NPI adoption and allowed us to estimate a less biased impact of the government's response over the GDP.
- The estimations of these economic impacts are relevant for future considerations on NPI's implementation

# THANK YOU

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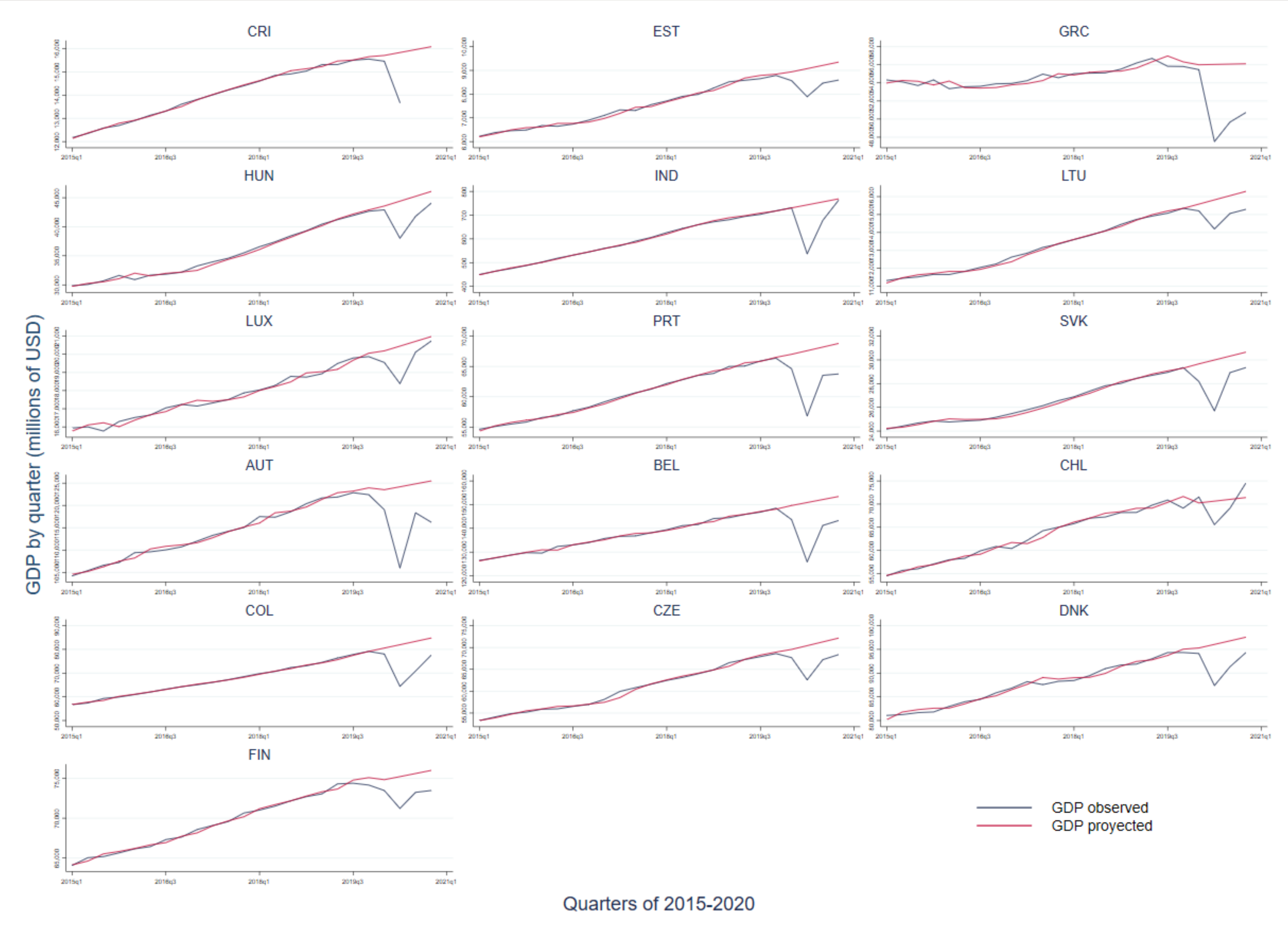
# ADDITIONAL SLIDES

# FORECASTED VS. ACTUAL GDP (1)

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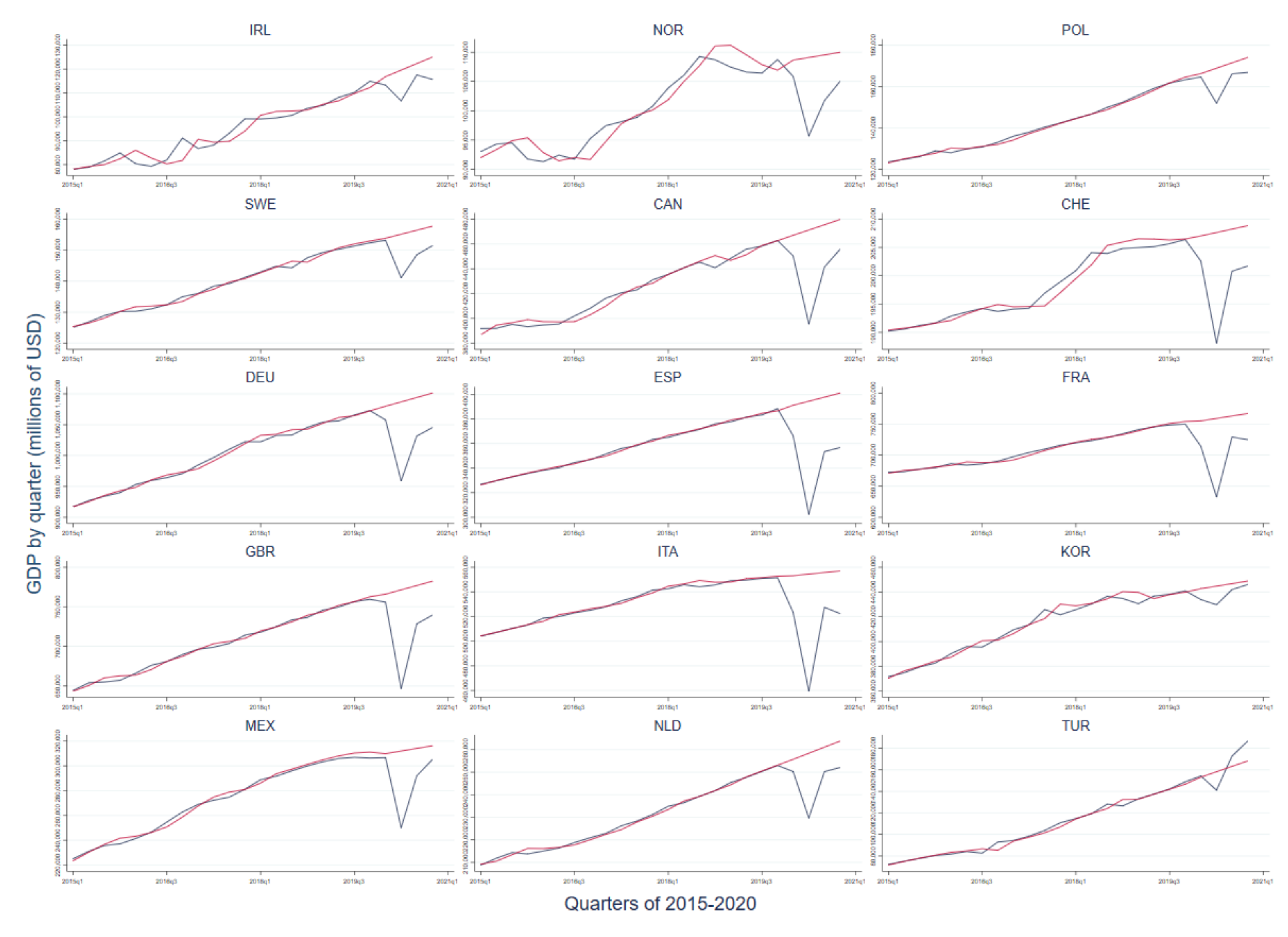
# FORECASTED VS. ACTUAL GDP (2)

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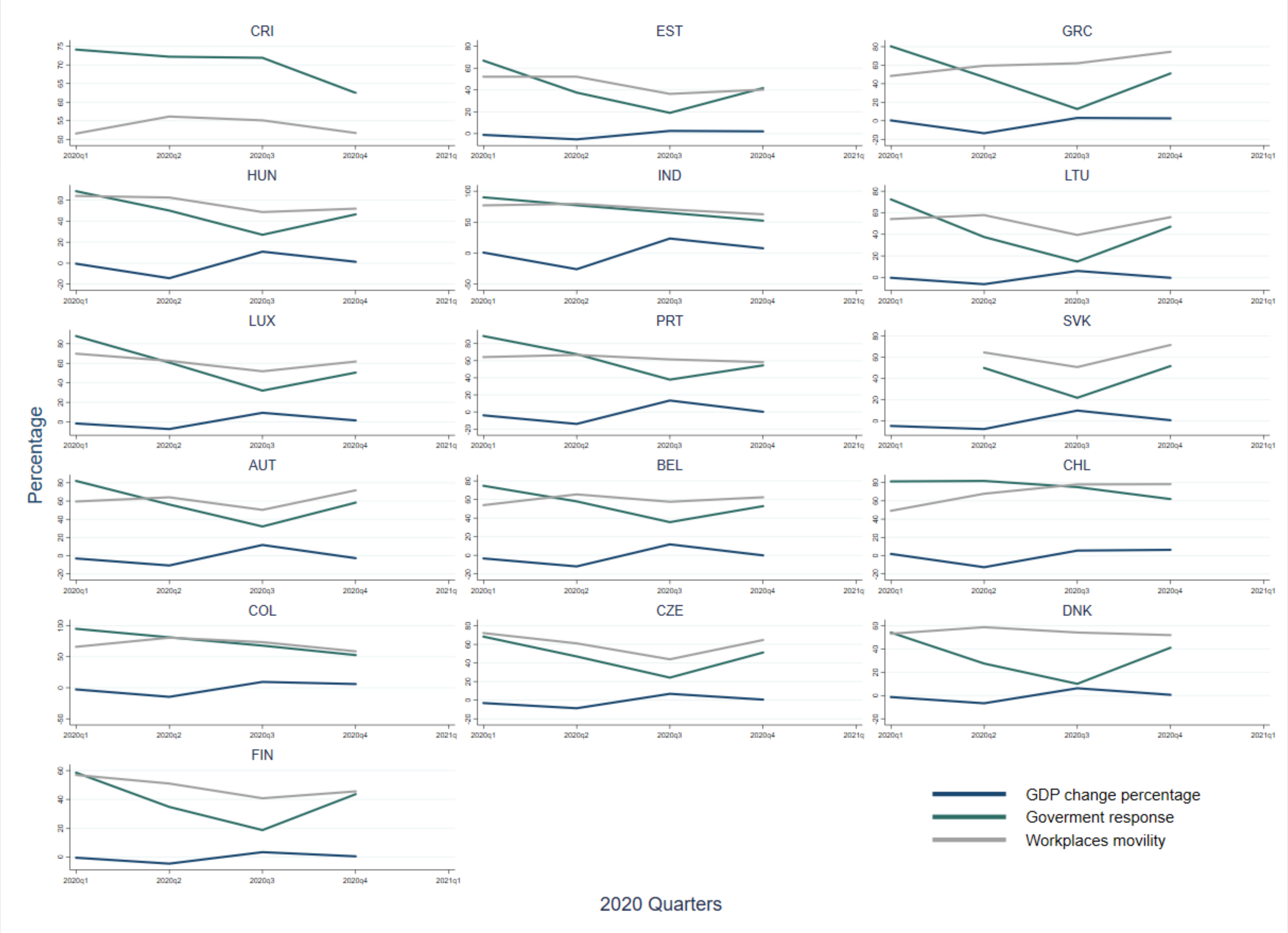
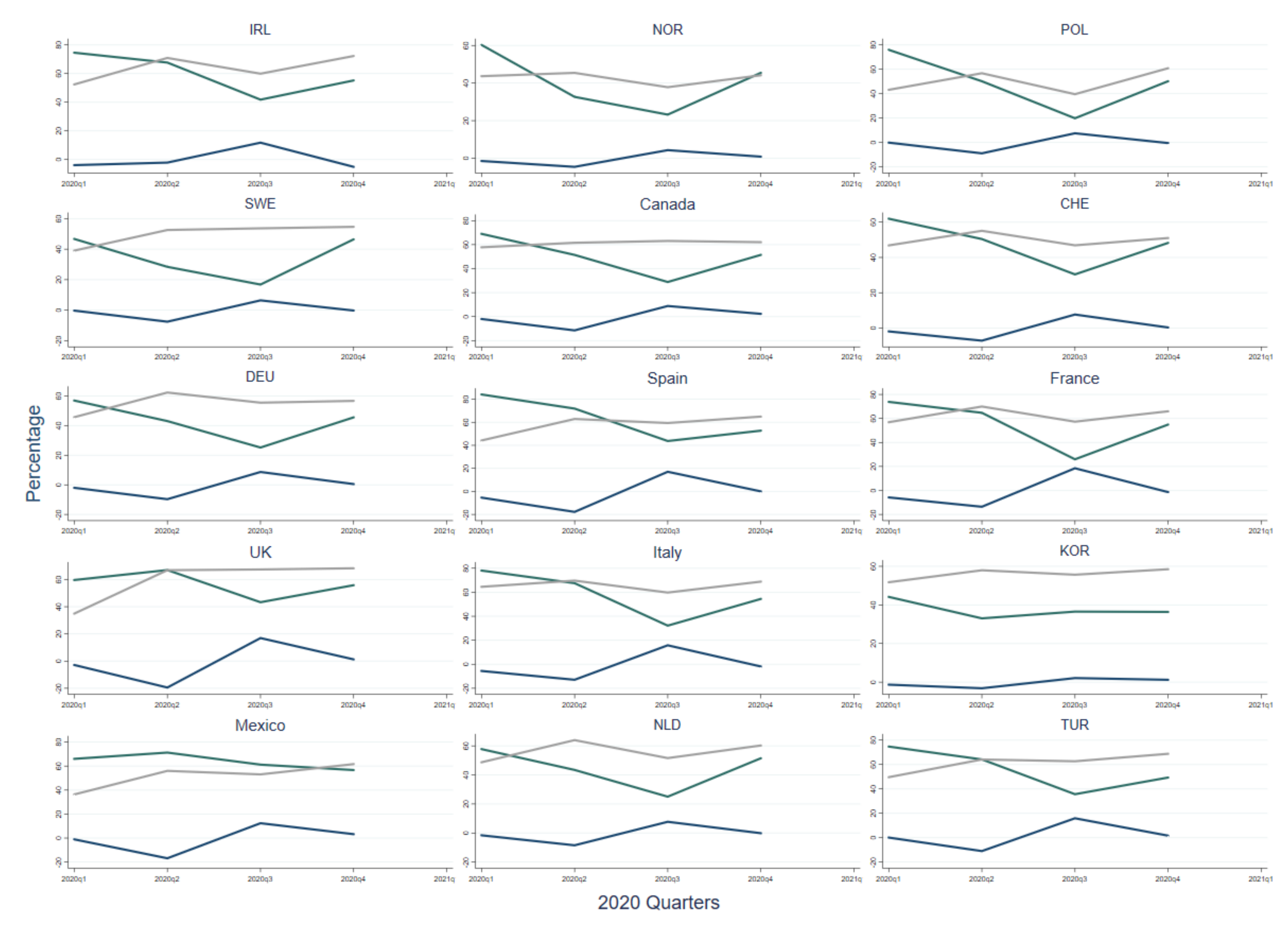
# GDP GROWTH, GOVERNMENT RESPONSE AND MOBILITY REDUCTION (1)



# GDP GROWTH, GOVERNMENT RESPONSE AND MOBILITY REDUCTION (2)

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